

Question				Marking details	Marks available																																			
					AO1	AO2	AO3	Total	Maths	Prac																														
4	(a)			<table><tr><th>Concentration of sodium thiosulfate (g / dm³)</th><th>Time 1 (s)</th><th>Time 2 (s)</th><th>Time 3 (s)</th><th>Mean time (s)</th></tr><tr><td>0.2</td><td>114</td><td>113</td><td>112</td><td>113</td></tr><tr><td>0.4</td><td>74</td><td>70</td><td>72</td><td>72</td></tr><tr><td>0.6</td><td>40</td><td>38</td><td>57</td><td>39</td></tr><tr><td>0.8</td><td>21</td><td>23</td><td>22</td><td>22</td></tr><tr><td>1.0</td><td>14</td><td>16</td><td>15</td><td>15</td></tr></table>	Concentration of sodium thiosulfate (g / dm ³)	Time 1 (s)	Time 2 (s)	Time 3 (s)	Mean time (s)	0.2	114	113	112	113	0.4	74	70	72	72	0.6	40	38	57	39	0.8	21	23	22	22	1.0	14	16	15	15			1	1	1	1
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				1.0	14	16	15	15																																
	(b)			award (2) for all points plotted correctly – tolerance $\pm\frac{1}{2}$ small square award (1) for 3 or 4 points plotted correctly award (1) for suitable curve		3		3	2																															
	(c)			the higher the concentration the higher the rate (1) because there are more particles in the same volume (1) therefore more chance of (successful) collisions / more (successful) collisions per second / greater frequency of collisions (1) or reverse argument the lower the concentration the lower the rate (1) because there are fewer particles in the same volume (1) therefore less chance of (successful) collisions / fewer (successful) collisions per second / lower frequency of collisions (1)	3			3																																

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	(d)			curve drawn below / to the left of plotted curve [must cover range of at least 0.5 g / dm ³ e.g. from 0.3 to 0.8]			1	1		
	(e)			award (2) for either of following 0.0063 6.3 × 10 ⁻³ award (1) for correct answer not given to 2 significant figures if answer incorrect award (1) for <i>M_r</i> 158 ecf possible from incorrect <i>M_r</i> e.g. 71		2		2	2	
				Question 4 total	3	5	2	10	5	1